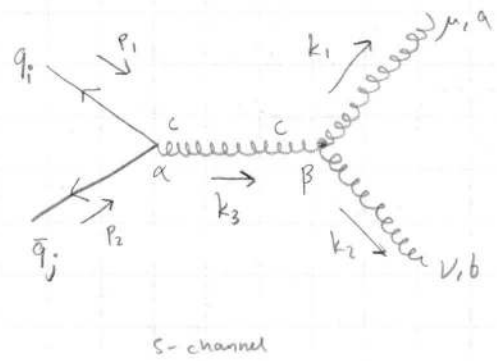
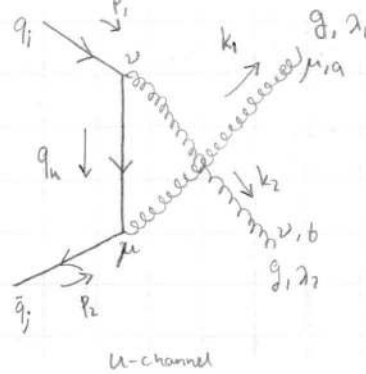
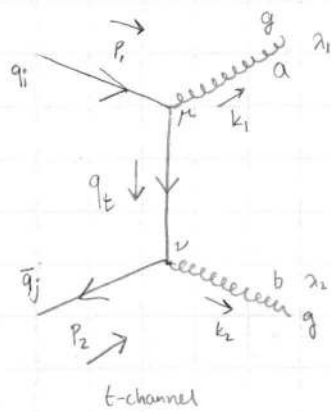


Exercise 4.8

$\bar{q}q \rightarrow gg$ scattering has the following Feynman-diagrams:



Here q_i is quark with colour i , $\bar{q}_j$ is antiquark with colour j . λ_1, λ_2 are polarization indices.

μ, ν, α, β are gluon Lorentz indices and a, b, c are gluon group indices. We have chosen not to show spin since it is not relevant.

$$q_t = p_i - k_1 = k_2 - p_j, \quad q_u = p_i - k_2 = k_1 - p_j, \quad q_s = p_i + p_j = k_1 + k_2$$

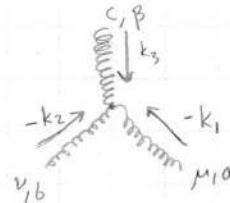
We use the following Feynman rules:

Fermion vertex:



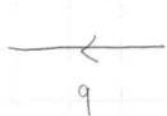
$$= ig \gamma^\mu T^a$$

Gluon-vertex:



$$= g f^{abc} \left[g^{\mu\nu} (-k_1 + k_2)^\beta + g^{\nu\beta} (-k_2 - k_3)^\mu + g^{\beta\mu} (k_3 + k_1)^\nu \right]$$

Fermion propagator:



$$= \frac{i}{q - m}$$

$$= \frac{i}{p_i - k_1 - m}$$

for $q = p_i - k_1$

External fermion:



$$= u_i(p), \text{ where } i \text{ is colour index.}$$

Gluon propagator in Feynman-gauge: $\mu \xrightarrow{k_3} \nu = -i \frac{g^{\mu\nu}}{k_3^2}$

External gluons: $\xleftarrow{k} = \epsilon_\mu^{\lambda_1}(k)$, $\xrightarrow{k} = \epsilon_\mu^{\lambda_2*}(k)$ for initial and final state respectively.

Using these we can compute the Amplitudes for M_{ttu} and M_s

$$i M_{ttu} = (ig)^2 \bar{v}_j(p_j) \left\{ \underbrace{\gamma^\nu T^b \frac{i}{p_i - k_1 - m} \gamma^\mu T^a}_{t\text{-channel part}} + \underbrace{\gamma^\mu T^a \frac{i}{k_1 - p_j - m} \gamma^\nu T^b}_{u\text{-channel part}} \right\} u_i(p_i) \epsilon_\mu^{\lambda_1*}(k_1) \epsilon_\nu^{\lambda_2}(k_2)$$

$i M_{ttu} \equiv M_{ttu}^{\mu\nu} \epsilon_\mu^{\lambda_1*}(k_1, \lambda_1) \epsilon_\nu^{\lambda_2}(k_2, \lambda_2)$, where we have defined $\epsilon_\mu^{\lambda_1*}(k_1) \equiv \epsilon_\mu^{\lambda_1*}(k_1)$ and

$$M_{ttu}^{\mu\nu} \equiv (ig)^2 \bar{v}_j(p_j) \left\{ \gamma^\nu T^b \frac{i}{p_i - k_1 - m} \gamma^\mu T^a + \gamma^\mu T^a \frac{i}{k_1 - p_j - m} \gamma^\nu T^b \right\} u_i(p_i)$$

The S-channel amplitude is

$$\begin{aligned}
 i\mathcal{M}_S &= ig \bar{V}_j(p_2) \gamma^\alpha T^c u_i(p_1) \frac{-ig^{\alpha\beta}}{k_3^2} \cdot gf^{abc} [g^{\mu\nu}(k_2-k_1)^\beta - g^{\nu\beta}(k_2+k_3)^\mu + g^{\beta\mu}(k_3+k_1)^\nu] \epsilon_\mu^{\lambda_1}(k_1) \epsilon_\nu^{\lambda_2}(k_2) \\
 &= -g^2 \bar{V}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} (g^{\mu\nu} [k_2-k_1]^\beta - g^{\nu\beta} [k_2+k_3]^\mu + g^{\beta\mu} [k_3+k_1]^\nu) \epsilon_\mu^*(k_1, \lambda_1) \epsilon_\nu^*(k_2, \lambda_2) \\
 &= \mathcal{M}_S^{\mu\nu} \epsilon_\mu^*(k_1, \lambda_1) \epsilon_\nu^*(k_2, \lambda_2)
 \end{aligned}$$

The total amplitude is $i\mathcal{M} = i\mathcal{M}_{\text{t-channel}} + i\mathcal{M}_S$:

$$\begin{aligned}
 i\mathcal{M} &= \left[(ig)^2 \bar{V}_j(p_2) \left\{ \gamma^\nu T^b \frac{i}{p_1 - p_2 - m} \gamma^\mu T^a + \gamma^\mu T^a \frac{i}{k_1 - p_2 - m} \gamma^\nu T^b \right\} u_i(p_1) + g^2 \bar{V}_j(p_2) \gamma_\beta T^c u_i(p_1) \right. \\
 &\quad \left. \frac{f^{abc}}{k_3^2} [g^{\mu\nu} [k_2-k_1]^\beta - g^{\nu\beta} [k_2+k_3]^\mu + g^{\beta\mu} [k_3+k_1]^\nu] \right] \epsilon_\mu^*(k_1, \lambda_1) \epsilon_\nu^*(k_2, \lambda_2)
 \end{aligned}$$

$$i\mathcal{M} \equiv \mathcal{M}^{\mu\nu} \epsilon_\mu^*(k_1, \lambda_1) \epsilon_\nu^*(k_2, \lambda_2)$$

Exercise 4.9

Show that $\sum_{\text{non-phys}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 \neq 0$

The polarization sum for physical polarization is

$$\sum_{i=1}^2 \epsilon_\mu(k, i) \epsilon_\nu^*(k, i) = -g_{\mu\nu} + \frac{\bar{k}_\mu k_\nu + k_\mu \bar{k}_\nu}{k \cdot \bar{k}} \equiv P_{\mu\nu}(k) \quad (1)$$

The polarization sum for all states is

$$\sum_{\lambda=0}^3 \epsilon_\mu(k, \lambda) \epsilon_\nu^*(k, \lambda) = -g_{\mu\nu} \quad (2)$$

Using (1) we get $\sum_{\text{phys}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 = \sum_{\lambda=0}^2 (\mathcal{M}^{\mu\nu})^* \mathcal{M}^{\alpha\beta} \epsilon_\mu(k_1, \lambda_1) \epsilon_\nu(k_2, \lambda_2) \epsilon_\alpha^*(k_1, \lambda_1) \epsilon_\beta^*(k_2, \lambda_2)$

$$\Rightarrow \sum_{\text{phys}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 = (\mathcal{M}^{\mu\nu})^* (\mathcal{M}^{\alpha\beta}) P_{\mu\alpha}(k_1) P_{\nu\beta}(k_2)$$

And using (2) we get $\sum_{\text{all states}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 = (\mathcal{M}^{\mu\nu})^* (\mathcal{M}^{\alpha\beta}) (-g_{\mu\alpha}) (-g_{\nu\beta})$

Using these expressions we get expression for non-physical sum:

$$\begin{aligned}
 \sum_{\text{non-phys}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 &= \sum_{\text{all states}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 - \sum_{\text{phys}} |\epsilon_1^\mu \epsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 \\
 &= (\mathcal{M}^{\mu\nu})^* (\mathcal{M}^{\alpha\beta}) [g_{\mu\alpha} g_{\nu\beta} - P_{\mu\alpha}(k_1) P_{\nu\beta}(k_2)] \equiv S
 \end{aligned}$$

$$\Rightarrow S = (M^{\mu\nu})^* (M^{\alpha\beta}) \left[g_{\mu\alpha} g_{\nu\beta} - \left(-g_{\mu\alpha} + \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu} \bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} \right) \left(-g_{\nu\beta} + \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu} \bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2} \right) \right]$$

$$S = (M^{\mu\nu})^* M^{\alpha\beta} \left[g_{\mu\alpha} \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu} \bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2} + g_{\nu\beta} \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu} \bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} - \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu} \bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu} \bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2} \right]$$

Now Let's calculate the contractions $M^{\mu\nu} k_{1\mu}$ and $M^{\mu\nu} k_{2\nu}$

$$k_{1\mu} M^{\mu\nu} = k_{1\mu} [M_{tm}^{\mu\nu} + M_s^{\mu\nu}] \quad , \quad \text{Let's start with } M_{tm}^{\mu\nu} k_{1\mu}$$

$$k_{1\mu} M_{tm}^{\mu\nu} = i (ig)^2 \bar{v}_j(p_2) \left\{ -\gamma^\nu T^b \frac{-\not{k}_1}{\not{p}_1 - \not{k}_1 - m} T^a + T^a \frac{\not{k}_1}{\not{k}_1 - \not{p}_2 - m} \gamma^\nu T^b \right\} u_i(p_1)$$

We now use Dirac equation: $(\not{p}_1 - m) u(p_1) = 0$ and $\bar{v}(p_2) (-\not{p}_2 - m) = 0$ to insert $(\not{p}_1 - m)$ and $(-\not{p}_2 - m)$ to the numerators via summation since $\not{k}_1 + 0 = \not{k}_1$

$$\Rightarrow k_{1\mu} M_{tm}^{\mu\nu} = i (ig)^2 \bar{v}_j(p_2) \left\{ -\gamma^\nu T^b \underbrace{\frac{\not{p}_1 - \not{k}_1 - m}{\not{p}_1 - \not{k}_1 - m}}_{=1} T^a + T^a \underbrace{\frac{\not{k}_1 - \not{p}_2 - m}{\not{k}_1 - \not{p}_2 - m}}_{=1} \gamma^\nu T^b \right\} u_i(p_1)$$

$$k_{1\mu} M_{tm}^{\mu\nu} = i (ig)^2 \bar{v}_j(p_2) \left\{ -\gamma^\nu T^b T^a + T^a T^b \gamma^\nu \right\} u_i(p_1) \quad \parallel \quad T^a T^b - T^b T^a = [T^a, T^b] = i f^{abc} T^c$$

$$= (ig)^2 \bar{v}_j(p_2) \left\{ i^2 f^{abc} T^c \gamma^\nu \right\} u_i(p_1) \quad \parallel \quad \text{Colour: } \bar{v}_j = \bar{v} \otimes \hat{c}_j$$

$$k_{1\mu} M_{tm}^{\mu\nu} = f^{abc} g^2 \bar{v}(p_2) \gamma^\nu u(p_1) T_{ji}^c$$

$$\text{and } g T^a c_i = T^a_{ij}$$

Now for the s-channel term

$$k_{1\mu} M_s^{\mu\nu} = g^2 \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} k_{1\mu} [g^{\mu\nu} (k_2 - k_1)^\beta - g^{\nu\beta} (k_2 + k_3)^\mu + g^{\beta\mu} (k_3 + k_1)^\nu]$$

$$= g^2 \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} [k_1^\nu (k_2 - k_1)^\beta - g^{\nu\beta} (k_2 + k_3) \cdot k_1 + k_1^\beta (k_3 + k_1)^\nu]$$

$$(k_2 + k_3) \cdot k_1 = (k_2 + k_3)(k_3 - k_2) = \stackrel{=0}{k_2^2} + k_3^2 = k_3^2 \quad \parallel \quad k_2^2 = 0 \quad \text{for on-shell gluon}$$

$$k_1^\nu (k_2 - k_1)^\beta + k_1^\beta (k_3 + k_1)^\nu = k_1^\nu k_2^\beta - k_1^\nu k_1^\beta + k_1^\beta k_3^\nu + k_1^\beta k_1^\nu = k_1^\nu k_2^\beta + k_1^\beta k_3^\nu \quad \parallel \quad k_1^\nu = (k_3 - k_2)^\nu$$

$$= k_3^\nu (k_2 + k_1)^\beta - k_2^\nu k_2^\beta = k_3^\nu k_3^\beta - k_2^\nu k_2^\beta$$

$$\Rightarrow k_{1\mu} M_s^{\mu\nu} = g^2 \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} [k_3^\nu k_3^\beta - k_2^\nu k_2^\beta - g^{\nu\beta} k_3^2]$$

The term $k_3^\beta k_3^\nu$ vanishes when contracted with the fermion current so we are left with

$$k_{1\mu} M_s^{\mu\nu} = -g^2 \bar{v}(p_2) \gamma^\nu u(p_1) T_{ji}^c f^{abc} - k_2^\nu g^2 \bar{v}(p_2) \not{k}_2 u(p_1) T_{ji}^c \frac{f^{abc}}{k_3^2}$$

Now we add the result and get:

$$k_{1\mu} M^{\mu\nu} = k_{1\mu} M_{t,u}^{\mu\nu} + k_{1\mu} M_s^{\mu\nu} = -k_2^\nu \frac{g^2 f^{abc}}{k_3^2} T_{ji}^c \bar{V}(p_2) \not{k}_2 u(p_1) \quad \parallel \quad k_3^2 = k_1^2 + k_2^2 + 2k_1 \cdot k_2$$

$$k_{1\mu} M^{\mu\nu} = -k_2^\nu \frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{V}(p_2) \not{k}_2 u(p_1) \equiv k_2^\nu A(k_1)$$

Similarly we get

$$k_{2\nu} M_{t,u}^{\mu\nu} = i(g)^2 \bar{V}_j(p_2) \left\{ T^b \frac{k_2}{k_2 - p_2 - m} \gamma^{\mu T^a} + \gamma^{\mu T^a} \frac{k_2}{p_1 - k_2 - m} T^b \right\} u(p_1)$$

We have used momentum conservation to get the denominators in terms of k_2

t-channel: $p_1 - k_1 = k_2 - p_2$, u-channel: $p_1 - k_2 = k_1 - p_2$

Now we use Dirac equation like previously:

$$k_{2\nu} M_{t,u}^{\mu\nu} = i(g)^2 \bar{V}_j(p_2) \left\{ T^b \gamma^{\mu T^a} - \gamma^{\mu T^a} T^b \right\} u(p_1)$$

$$= i(g)^2 \bar{V}_j(p_2) \left\{ -i f^{abc} T^c \gamma^\mu \right\} u(p_1) = -g^2 f^{abc} T_{ji}^c \bar{V}(p_2) \gamma^\mu u(p_1)$$

And for the s-channel

$$k_{2\nu} M_s^{\mu\nu} = g^2 \bar{V}_j(p_2) \gamma_\beta T^c u(p_1) \frac{f^{abc}}{k_3^2} \left(g^{\mu\nu} (k_2 - k_1)^\beta - g^{\nu\beta} (k_2 + k_3)^\mu + g^{\beta\mu} (k_3 + k_1)^\nu \right) k_{2\nu}$$

$$= g^2 T_{ji}^c \bar{V}(p_2) \gamma_\beta u(p_1) \frac{f^{abc}}{k_3^2} \left(k_2^\mu (k_2 - k_1)^\beta - k_2^\beta (k_2 + k_3)^\mu + g^{\beta\mu} (k_3 + k_1) \cdot k_2 \right)$$

$$(k_3 + k_1) \cdot k_2 = (k_3 + k_1)(k_3 - k_1) = k_3^2, \quad k_2^\mu (k_2 - k_1)^\beta - k_2^\beta (k_2 + k_3)^\mu = -k_2^\mu k_1^\beta - k_2^\beta k_3^\mu$$

$$= -(k_3^\mu - k_1^\mu) k_1^\beta - k_2^\beta k_3^\mu = -k_3^\mu k_3^\beta - k_1^\mu k_1^\beta \quad \parallel$$

$$\Rightarrow k_{2\nu} M_s^{\mu\nu} = g^2 T_{ji}^c \bar{V}(p_2) \gamma^\mu u(p_1) f^{abc} - g^2 T_{ji}^c f^{abc} \bar{V}(p_2) \gamma_\beta u(p_1) \left[k_3^\mu k_3^\beta + k_1^\mu k_1^\beta \right] \cdot \frac{1}{k_3^2}$$

As before $k_3^\mu k_3^\beta$ vanishes when contracted by fermion current and $k_3^2 = 2k_1 \cdot k_2$

$$\Rightarrow k_{2\nu} M^{\mu\nu} = k_{2\nu} M_{t,u}^{\mu\nu} + k_{2\nu} M_s^{\mu\nu} = -k_1^\mu \frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{V}(p_2) \not{k}_1 u(p_1)$$

$$k_{2\nu} M^{\mu\nu} \equiv k_1^\mu A(k_1)$$

$$T^{c\dagger} = T^c \Rightarrow T_{ij}^{c\dagger} = T_{ji}^c$$

$$\text{We will also use relation } k_{2\nu} (M^{\mu\nu})^* = k_1^\mu A^* = -k_1^\mu \frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{u}(p_1) \not{k}_1 v(p_2)$$

since $(\psi^\mu)^*$ affects only spinor and colour part of which we take Hermitian adjoint of

$$\bar{k}_{2\nu} = R k_{2\nu} M^{\mu\nu} = R k_1^\mu A = \bar{k}_1^\mu A, \quad \text{where } R \text{ is operator that turns } k \rightarrow \bar{k}$$

$$\Rightarrow \bar{k}_{2\nu} = \bar{k}_1^\mu A(k_2)$$

Now we use the Relations $k_{\mu} M^{\mu\nu} = k_2^{\nu} A(k_1)$ etc to simplify our S:

$$S = (M^{\mu\nu})^* M^{\alpha\beta} \left[g_{\mu\alpha} \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu} \bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2} + g_{\nu\beta} \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu} \bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} - \left(\frac{\bar{k}_{1\mu} k_{1\mu} + k_{1\mu} \bar{k}_{1\mu}}{k_1 \cdot \bar{k}_1} \right) \left(\frac{\bar{k}_{2\nu} k_{2\nu} + k_{2\nu} \bar{k}_{2\nu}}{k_2 \cdot \bar{k}_2} \right) \right]$$

Let's calculate term by term

$$\begin{aligned} \frac{1}{k_2 \cdot \bar{k}_2} (M^{\mu\nu})^* M^{\alpha\beta} g_{\mu\alpha} (\bar{k}_{2\nu} k_{2\beta} + k_{2\nu} \bar{k}_{2\beta}) &= \frac{g_{\mu\alpha}}{k_2 \cdot \bar{k}_2} [\bar{k}_1^{\mu} A^*(k_2) k_{2\beta} + k_1^{\mu} A^*(k_2) \bar{k}_{2\beta}] M^{\alpha\beta} \\ &= \frac{1}{k_1 \cdot \bar{k}_1} [\bar{k}_{1\alpha} k_{2\beta} A^*(k_2) + k_{1\alpha} \bar{k}_{2\beta} A^*(k_2)] M^{\alpha\beta} \\ &= \frac{1}{k_2 \cdot \bar{k}_2} [\underbrace{\bar{k}_2^{\beta} k_{2\beta} + k_2^{\beta} \bar{k}_{2\beta}}_{= 2 k_2 \cdot \bar{k}_2}] A(k_1) A^*(k_2) = 2 A(k_1) A^*(k_2) \end{aligned}$$

Similarly we get

$$\frac{1}{k_1 \cdot \bar{k}_1} (M^{\mu\nu})^* M^{\alpha\beta} [\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu} \bar{k}_{1\alpha}] = 2 A(k_2) A^*(k_1)$$

And

$$\begin{aligned} (M^{\mu\nu})^* M^{\alpha\beta} \left[\frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu} \bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} \right] \left[\frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu} \bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2} \right] &= \frac{(M^{\mu\nu})^* M^{\alpha\beta}}{(k_1 \cdot \bar{k}_1)(k_2 \cdot \bar{k}_2)} \left[\underbrace{\bar{k}_{1\mu} k_{1\alpha} \bar{k}_{2\nu} k_{2\beta} + \bar{k}_{1\mu} k_{1\alpha} k_{2\nu} \bar{k}_{2\beta}}_{= 0} \right. \\ &\quad \left. + k_{1\mu} \bar{k}_{1\alpha} \bar{k}_{2\nu} k_{2\beta} + k_{1\mu} \bar{k}_{1\alpha} k_{2\nu} \bar{k}_{2\beta} \right] \\ &= \frac{1}{(k_1 \cdot \bar{k}_1)(k_2 \cdot \bar{k}_2)} \left[\bar{k}_2^{\nu} A^*(k_1) k_{1\alpha} k_{2\nu} \bar{k}_1^{\beta} A(k_2) + k_{1\mu} \bar{k}_2^{\beta} A(k_1) \bar{k}_{1\mu} A^*(k_2) k_{2\beta} \right] \\ &= A(k_1) A^*(k_2) + A(k_2) A^*(k_1) \end{aligned}$$

$$\Rightarrow S = 2 A(k_1) A^*(k_2) + 2 A(k_2) A^*(k_1) - (A(k_1) A^*(k_2) + A(k_2) A^*(k_1))$$

$$S = A(k_1) A^*(k_2) + A(k_2) A^*(k_1), \text{ Now that we have simplified this}$$

$$\text{Let's expand again: with } A(k_1) = -\frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{v}(p_2) \not{k}_2 u(p_1), \quad A(k_2) = -\frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{v}(p_2) \not{k}_1 u(p_1)$$

$$\text{and } A^*(k_1) = -\frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{u}(p_1) \not{k}_2 v(p_2), \quad A^*(k_2) = -\frac{g^2 f^{abc}}{2k_1 \cdot k_2} T_{ji}^c \bar{u}(p_1) \not{k}_1 v(p_2)$$

$$\Rightarrow A(k_1) A^*(k_2) = \frac{g^4 f^{abc}}{4(k_1 \cdot k_2)^2} f^{mno} (T_{ji}^c T_{ji}^o) \bar{v}(p_2) \not{k}_2 u(p_1) \bar{u}(p_1) \not{k}_1 v(p_2)$$

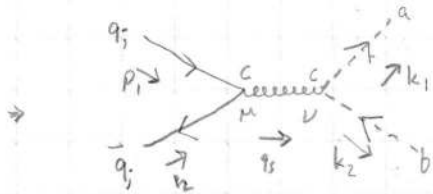
$$\Rightarrow A(k_2) A^*(k_1) = \frac{g^4 f^{abc}}{4(k_1 \cdot k_2)^2} f^{mno} (T_{ji}^c T_{ji}^o) \bar{v}(p_2) \not{k}_1 u(p_1) \bar{u}(p_1) \not{k}_2 v(p_2)$$

$$\Rightarrow S = -\frac{g^4 f^{abc}}{4(k_1 \cdot k_2)^2} f^{mno} (T^c)_{ji} (T^o)_{ji} \bar{v}(p_2) [k_2 u(p_1) \bar{u}(p_1) k_1 + k_1 u(p_1) \bar{u}(p_1) k_2] v(p_2) \neq 0$$

$\Rightarrow$ We have shown that $\sum_{\text{non-phys}} |\epsilon_1^\mu \epsilon_2^\nu M_{\mu\nu}|^2 \neq 0$

Exercise 4/10

For $q\bar{q} \rightarrow c\bar{c}$ we only have s-diagram since there is no direct interaction/coupling of q and c fields. The Feynman diagram is then



The ghost vertex gives the contribution: $-g f^{acb} k_1^\nu = g f^{abc} k_1^\nu$ since f^{abc} is totally antisymmetric.

We note that the vertex depends on the momentum of ghost-particle so we actually get two contribution from the s-channel: one where ghost has momentum of k_2 and one where it has momentum of k_1 . Because of this our Matrix element looks like the following: (External legs of scalar field is 1)

$$i M_{\text{ghost}} = \bar{v}_j(p_2) [ig \gamma^\mu T^c] u_i(p_1) \left(-\frac{ig^{\mu\nu}}{q_s^2} \right) \cdot (g f^{abc} k_1^\nu + g f^{abc} k_2^\nu) \quad \left\| \begin{array}{l} q_s^2 = (k_1 + k_2)^2 \\ = -2k_1 \cdot k_2 \end{array} \right.$$

two different contributions from two different possible final state.

$$= -\frac{g^2}{2(k_1 \cdot k_2)} f^{abc} T_{ji}^c \bar{v}(p_2) [k_1 + k_2] u(p_1)$$

$$\Rightarrow |M_{\text{ghost}}|^2 = \frac{g^4}{4(k_1 \cdot k_2)^2} f^{abc} f^{mno} T_{ji}^c T_{ji}^o (\bar{v}(p_2) [k_1 + k_2] u(p_1)) (\bar{u}(p_1) [k_1 + k_2] v(p_2))$$

$$= \frac{g^4}{4(k_1 \cdot k_2)^2} f^{abc} f^{mno} T_{ji}^c T_{ji}^o \bar{v}(p_2) [k_1 u(p_1) \bar{u}(p_1) k_2 + k_2 u(p_1) \bar{u}(p_1) k_1] v(p_2)$$

The terms $\bar{v}(p_2) k_1 u(p_1) \bar{u}(p_1) k_1 v(p_2)$ will vanish because it will be proportional to $k_1^2 = 0$ when taking the trace so we set it 0, same with k_2 replaced by k_1

$$\Rightarrow |M_{\text{ghost}}|^2 = \frac{g^4}{4(k_1 \cdot k_2)^2} f^{abc} f^{mno} T_{ji}^c T_{ji}^o \bar{v}(p_2) [k_2 u(p_1) \bar{u}(p_1) k_1 + k_1 u(p_1) \bar{u}(p_1) k_2] v(p_2)$$

This term is exactly same as $S = \sum_{\text{non-phys}} |\epsilon_1^\mu \epsilon_2^\nu M_{\mu\nu}|^2$ but with opposite sign

$$\Rightarrow \sum_{\text{non-phys}} |\epsilon_1^\mu \epsilon_2^\nu M_{\mu\nu}|^2 + |M_{\text{ghost}}|^2 = 0 \quad \square$$

So $\sum_{\text{all}} |\epsilon_1^\mu \epsilon_2^\nu M_{\mu\nu}|^2 + |M_{\text{ghost}}|^2 = \sum_{\text{physical}} |\epsilon_1^\mu \epsilon_2^\nu M_{\mu\nu}|^2$